Spoken Tutorial on Your First AI Application

Module	Module 3: An introduction to Building
Episode	3.3: Hands-On Project: Your First AI Application
Learning Objective	At the end of this tutorial learner will be able to 1. Explain what an API is. 2. Identify how to get a Google AI Studio API key. 3. Apply Python code to interact with the Gemini API. 4. Generate marketing slogans using an AI model.
Approx. Duration	3-4 min
Outline	• Understanding APIs • Introduction to Gemini API • Getting your API Key • Setting up Google Colab • Installing and configuring the Gemini API • Initializing the AI model • Defining the product topic • Crafting effective prompts • Comparing prompt effectiveness • Running your AI application • Experimenting with challenges
Meta Tags	AI application, Gemini API, Python, Google Colab, API key, marketing slogans, generative AI, hands-on project, AI Essentials, coding
Pre-requisite Tutorial	Episode 1 and 2 of this Module

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on **Your First AI Application**.
Learning Objectives Slide	In this tutorial, you will learn to, • Explain what an **API** is. • Identify how to get a **Google AI Studio API key**. • Apply **Python** code to interact with the **Gemini API**. • Generate marketing slogans using an **AI** model.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have watched **Episode 1**. Also watch **Episode 2** of this **Module**.

Illustration showing an API as a waiter connecting a customer (application) to a kitchen (server)	What exactly is an **API**? Think of an **API**, or **Application Programming Interface**. It is like a waiter in a restaurant. You tell the waiter what you want. The waiter then communicates with the kitchen to get it. You don't need to know how the food is prepared. Similarly, an **API** connects your application to a server.
Illustration showing Gemini API as a specialized chef creating unique dishes like marketing slogans	Now, let's introduce the **Gemini API**. Continuing our analogy, the **Gemini API** is like a specialized chef. This chef can create unique dishes, such as marketing slogans. It does this based on your specific requests. It makes **Google's** powerful **AI** model accessible to developers.
Browser showing Google AI Studio interface with 'Get API key' button highlighted, user clicking it	First, navigate to **Google AI Studio**. **Sign in** with your **Google** account. **Click** on **Get API key**. This creates a new **API key**. Then, **copy** this **API key**. Remember to keep your **API key** secure.
Browser showing a new Google Colab notebook interface	Next, open a new **Google Colab** notebook. Do this directly in your browser. **Google Colab** provides a **Python** environment. Here you will write and run your code.
Screen close-up: user typing '!pip install google-generativeai' and 'genai.configure(api_key="YOUR_API_KEY")' in Google Colab	In **Colab**, first install the generative **AI** package. Use `!pip install` for this. Then, **import** and configure it. Use your copied **API key** with `genai.configure`.
Screen close-up: user typing 'model = genai.GenerativeModel("gemini-pro")' in Google Colab	Now, initialize the **gemini-pro** model. Create an instance of `genai.GenerativeModel`. This prepares the model. It will then receive your **prompts**.
Screen close-up: user typing 'product_topic = 'a new brand of eco-friendly sneakers'' in Google Colab	Let us define our product topic. Store it in a variable. For example, `product_topic = 'a new brand of eco-friendly sneakers'`. This will be the basis for our marketing slogans.
Screen close-up: user typing 'prompt = f"Create 5 catchy marketing slogans for {product_topic}"' in Google Colab	Now, let's craft an improved **prompt**. Make it clear and specific. For example, `prompt = f"Create 5 catchy marketing slogans for {product_topic}"`. This guides the **AI** model. It generates relevant output.

Side-by-side: vague prompt 'slogans for sneakers' output vs improved prompt 'Create 5 catchy marketing slogans for a new brand of eco-friendly sneakers' output	Let's compare **prompt** effectiveness. A vague **prompt**, like 'slogans for sneakers', gives generic results. But a specific **prompt** directs the **AI**. For example, a prompt can be crafted. Use 'Create 5 catchy marketing slogans for a new brand of eco-friendly sneakers'. This produces targeted and useful output.
Google Colab screen showing code block with play button, AI generating response below	Now, run your **AI** application. **Press** the **play button** next to your code block in **Colab**. The **AI** processes your **prompt**. It will **print** the generated marketing slogans. This happens in a few seconds.
Screen close-up: user modifying 'product_topic' variable in Google Colab, with a thought bubble indicating experimentation	Here is a challenge for you. Modify the `product_topic` variable. Explore different products, like 'a new spicy hot sauce'. Or try 'a productivity app'. Observe the **AI's** varied slogan creations.
Summary Slide	Let us summarize what we learned. We learned about **APIs**. We obtained an **API key**. We set up **Google Colab**. Finally, we built our first **AI** application. This generates marketing slogans using the **Gemini API**.
Assignment Slide	Now, for an assignment, experiment with different product topics. Also, try various **prompt** structures. Generate diverse marketing slogans. Try to add constraints to your **prompt**. For example, add constraints like 'slogans under 10 words'. Compare the results.
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